

DEEP LEARNING APPROACHES FOR EFFICIENT AND ACCURATE DNA SEQUENCE ALIGNMENT USING LARGE LANGUAGE MODELS

THIS RESEARCH INTRODUCES A HYBRID DEEP LEARNING PIPELINE LEVERAGING DNABERT TO OVERCOME THE LIMITATIONS OF TRADITIONAL DNA ALIGNMENT. THE MODEL CAPTURES SEMANTIC AND CONTEXTUAL RELATIONSHIPS, ACHIEVING HIGH ACCURACY (AUROC > 0.90) EVEN IN SEQUENCES WITH LOW IDENTITY. THE RESULTS CONFIRM THAT CONTEXTUAL EMBEDDINGS UNCOVER FUNCTIONAL SIMILARITIES INVISIBLE TO IDENTITY-BASED METRICS.

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INTRODUCTION

DNA SEQUENCE ALIGNMENT IS ESSENTIAL FOR DISCOVERING FUNCTIONAL AND EVOLUTIONARY SIMILARITIES. HOWEVER, TRADITIONAL ALGORITHMS LIKE BLAST RELY ON BASE-TO-BASE IDENTITY, OFTEN FAILING WHEN SEQUENCES ARE FUNCTIONALLY RELATED BUT STRUCTURALLY DIVERGENT DUE TO MUTATIONS. THIS STUDY INTEGRATES LARGE LANGUAGE MODELS (LLMS) TO REPRESENT DNA AS CONTEXTUAL EMBEDDINGS, MOVING BEYOND LITERAL MATCHING TO A DEEPER SEMANTIC UNDERSTANDING OF GENOMIC DATA.

OBJECTIVE

The research aims to integrate DNABERT for high-dimensional DNA representation and compare identity-based metrics against embedding-based similarity. A key goal is to validate the hybrid framework using Precision, Recall, F1-score, and AUROC across multiple datasets to ensure robustness.

METHODOLOGY

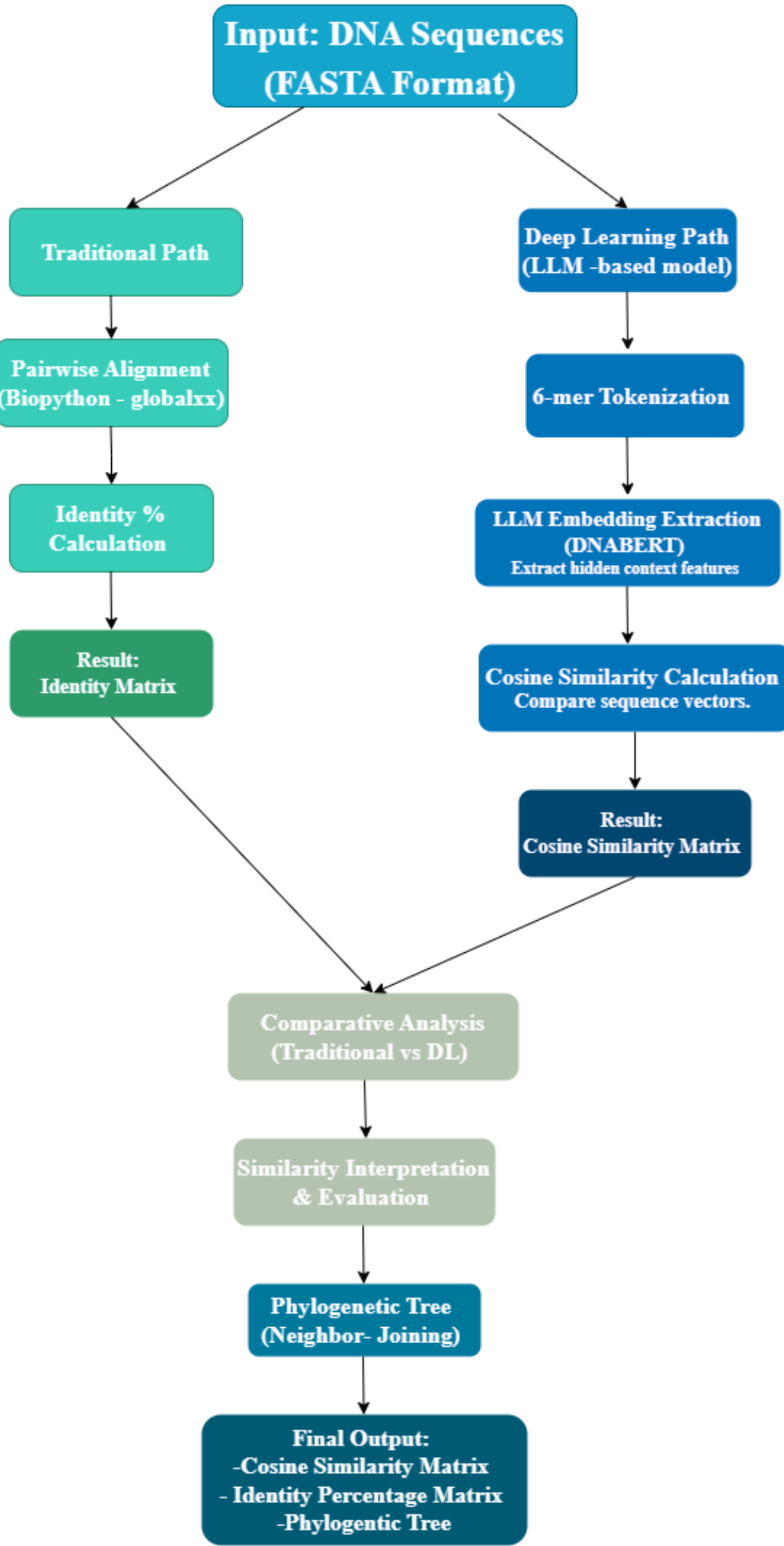
The framework analyzes DNA sequences using two parallel paths: a traditional alignment path to compute identity similarity, and a DNABERT-based path to extract embeddings and measure cosine similarity.

Both outputs are compared to capture direct and contextual relationships, followed by phylogenetic tree construction for evolutionary analysis.

RESULTS/FINDINGS

Experiments were conducted on two datasets (20 and 70 sequences). Visual results are presented for the 20-sequence dataset for clarity, while the larger dataset validates robustness.

- Traditional methods: low similarity (~24%)
- DNABERT: high similarity (~99%)
- Captures hidden relationships
- Stable across datasets
- AUROC up to 0.95

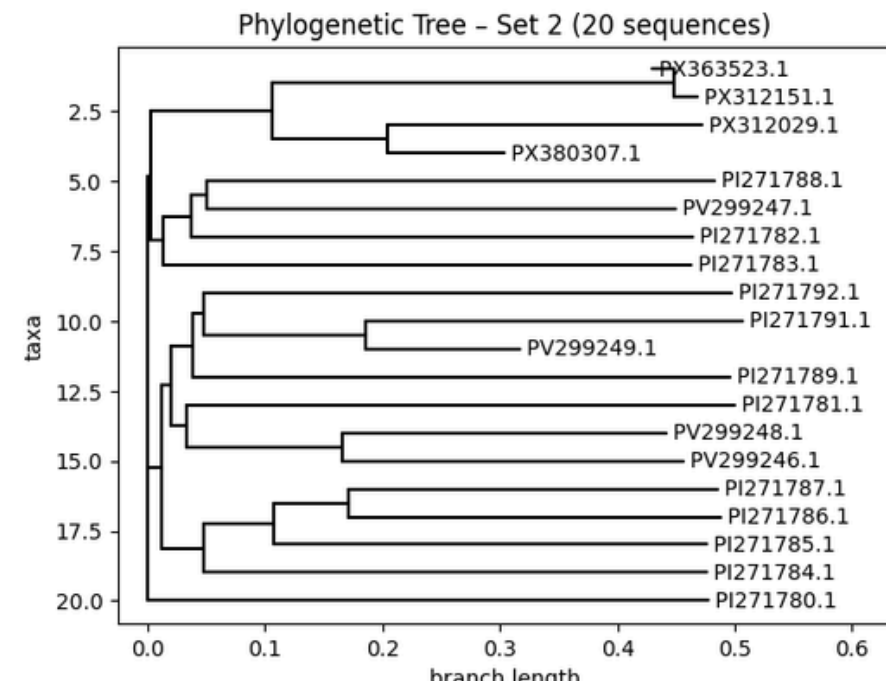


QUANTITATIVE EVALUATION

Dataset	Precision	Recall	F1-Score	AUROC
Set 1 (20 Seqs)	0.95	0.90	0.92	0.955
Set 2 (70 Seqs)	0.95	0.90	0.92	0.901

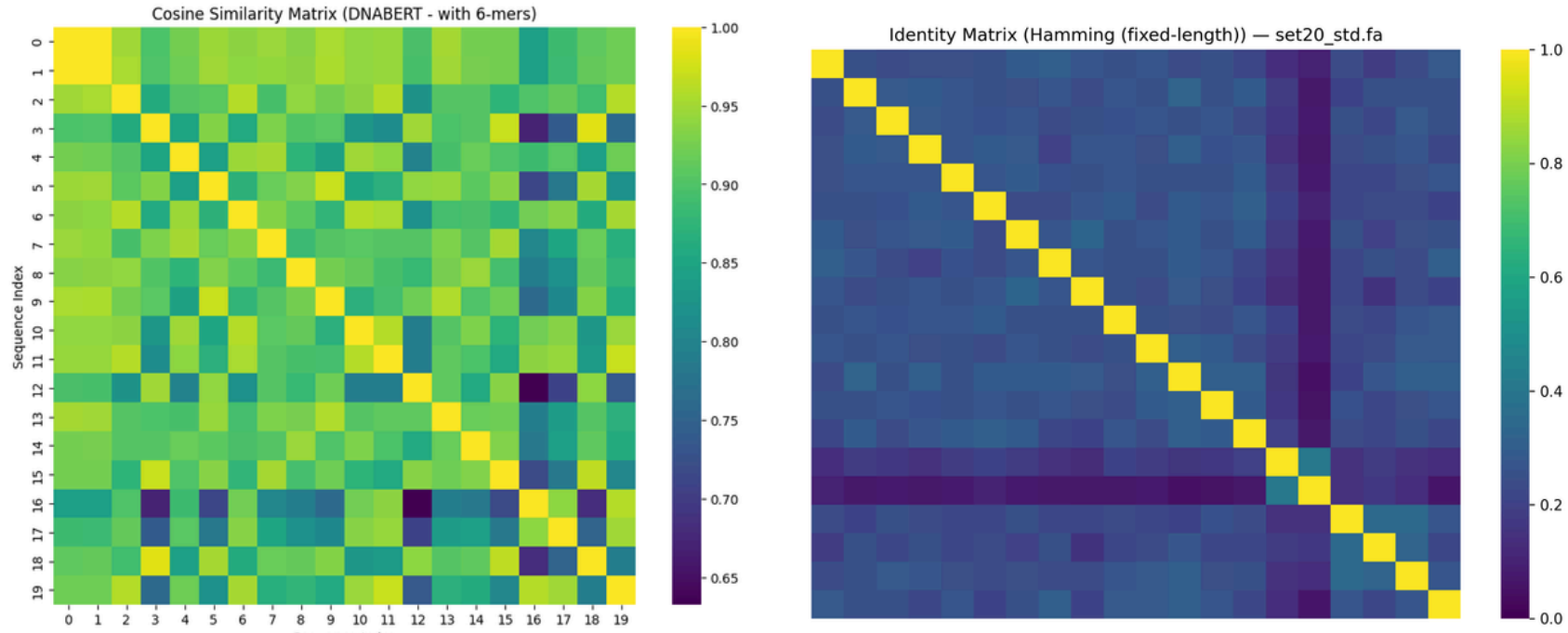
- AUROC: 0.90+
- Sequences: 20 & 70
- Similarity: 99%

PHYLOGENETIC TREE



- Shows clustering of similar DNA sequences
 - Reveals evolutionary relationships
 - Supports alignment and similarity results
- Sequences with higher similarity are grouped within the same clusters.

COMPARATIVE VISUALIZATION



DNABERT maintains high similarity (>98%) even when identity drops below 40%, highlighting its ability to capture hidden functional relationships.

Research Core:

Contextual embeddings reveal biological relationships that literal identity scores cannot.

Tools:

Biopython · ClustalW · DNABERT · PyTorch · scikit-learn · Google Colab.

Publications

- Deep Learning Approaches for Efficient and Accurate DNA Sequence Alignment Using Large Language Models

Comput. Sci. Math. Forum (ICETI 2026)

- Comparative Analysis of DNA Sequence Similarity Using Pairwise Alignment and DNABERT Embeddings

Under Review – IJACSA

CONCLUSION

The hybrid framework proves that LLM-based embeddings are a robust alternative for identifying hidden evolutionary relationships. Future work will focus on integrating newer models like DNABERT-2 and applying the framework to larger multi-species datasets.